

ECON 5260 Problem Set 4 Solution

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Full mark: 60 points.

Question 1 (24 points)

1. (6 points) The labour supply curve can be derived from the household's FOC regarding N_t :

$$\chi_t N_t^\eta = \frac{W_t}{P_t} C_t^{-\sigma}. \quad \text{.....} \quad (2 \text{ points})$$

In the equilibrium, $N_t = Y_t/Z_t$, $C_t = Y_t$. Thus, last equation implies that

$$\frac{W_t}{P_t} = Z_t^\sigma \chi_t N_t^{\eta+\sigma}, \quad \text{.....} \quad (4 \text{ points})$$

which describes the labour supply curve. A positive taste shock χ_t will decrease the labour supply N_t by shift the curve upwardly. (6 points)

2. (4 points) In the flexible-price equilibrium, $W_t/P_t = Z_t/\mu$ where $\mu > 1$ is the markup. From the labour supply curve, we obtain

$$Y_t = \mu^{-\frac{1}{\eta+\sigma}} Z_t^{\frac{1+\eta}{\eta+\sigma}} \chi_t^{-\frac{1}{\eta+\sigma}}.$$

Log-linearization yields

$$\hat{y}_t^f = \frac{1+\eta}{\eta+\sigma} \hat{z}_t - \frac{1}{\eta+\sigma} \hat{\chi}_t.$$

3. (4 points) The taste shock negatively affects labour and thus output (consumption) in the flexible-price equilibrium, since it reduces the incentives of households to provide labour.

4. (10 points) The optimal decision of purchasing bonds yields

$$Y_t^{-\sigma} = \beta(1 + i_t) \mathbb{E}_t \frac{P_t}{P_{t+1}} Y_{t+1}^{-\sigma}.$$

Log-linearization yields:

$$\hat{y}_t = \mathbb{E}_t \hat{y}_{t+1} + \frac{1}{\sigma} [\mathbb{E}_t \pi_{t+1} - i_t].$$

Define $x_t = \hat{y}_t - \hat{y}_t^f$. Then we can rewrite the last equation as

$$x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\sigma} [i_t - \mathbb{E}_t \pi_{t+1} - r_t^n],$$

where $r_t^n = -\sigma(\hat{y}_t^f - \mathbb{E}_t \hat{y}_{t+1}^f)$. (4 points) Using the flexible-price equilibrium expression for y_t^f , we can further express r_t^n as

$$r_t^n = \sigma \frac{1 + \eta}{\eta + \sigma} (\mathbb{E}_t \hat{z}_{t+1} - \hat{z}_t) - \frac{\sigma}{\eta + \sigma} (\mathbb{E}_t \hat{\chi}_{t+1} - \hat{\chi}_t). \quad \dots\dots (6 \text{ points})$$

The flexible-price real interest rate r_t^n negatively depends on the expected change of $\hat{\chi}_t$. The intuition is that expected increasing in $\hat{\chi}_{t+1}$ with respect to $\hat{\chi}_t$ implies that current consumption or output is greater than the value in the next period, i.e., marginal utility for one unit extra consumption is greater in the next period. This makes household become more patient (equivalently, the stochastic discounting factor increases), which implies a lower real interest rate. (10 points)

Question 2 (14 points)

The FOC is now

$$\chi_t N_t^\eta = \psi_t \frac{W_t}{P_t} C_t^{-\sigma},$$

with $N_t = Y_t/Z_t$, $C_t = Y_t$. Thus, we have

$$\frac{W_t}{P_t} = Z_t^\sigma \frac{\chi_t}{\psi_t} N_t^{\eta+\sigma}. \quad \dots\dots (4 \text{ points})$$

1. A positive ψ_t is like a negative χ_t . It increases the labour supply due to this separable preference. (6 points)
2. Similarly,

$$\hat{y}_t^f = \frac{1 + \eta}{\eta + \sigma} \hat{z}_t - \frac{1}{\eta + \sigma} (\hat{\chi}_t - \hat{\psi}_t).$$

The intuition is similar. (9 points)

3. The real marginal cost is

$$\hat{\varphi}_t = \eta(\hat{y}_t - \hat{z}_t) + \sigma\hat{y}_t - \hat{z}_t + \hat{\chi}_t - \hat{\psi}_t. \quad \dots\dots\dots (11 \text{ points})$$

The last two terms are taste shocks. Following the derivation of (8.26), we have

$$\pi_t = \beta\mathbb{E}_t\pi_{t+1} + \kappa x_t + \kappa(\hat{\chi}_t - \hat{\psi}_t). \quad \dots\dots\dots (14 \text{ points})$$

Question 3 (10 points)

1. (6 points) A small $\phi_\pi \leq 1$ implies the nominal interest rate does not respond to the inflation sufficiently. If people expect the inflation will increase, the real interest rate $i_t - \mathbb{E}_t\pi_{t+1}$ may decrease. Consequently, demand increases and output goes up. According to the AS curve, the increasing x_t will in turn push up the inflation rate. Therefore, the expected increasing inflation can be fulfilled. (4 points)

Technically, the eigenvalues of coefficient matrix of $\begin{bmatrix} x_t \\ \pi_t \end{bmatrix}$ are smaller than 1, indicating that the system has sink solution instead of saddle solution. (2 points)

2. (4 points) When $\phi_x > 0$, the interest rate responds more sufficiently than the case in (a). That is, the monetary policy is more counter-cyclical. As a result, lower value of ϕ_π is required to ensure the local determinacy (uniqueness of solution).

Question 4 (12 points)

1. (9 points) Substituting $\pi_t = \gamma e_t$ into the NKPC, we obtain

$$\gamma e_t = \beta\rho\gamma e_t + \kappa x_t + e_t.$$

This implies that

$$x_t = \underbrace{\frac{\gamma(1 - \beta\rho) - 1}{\kappa}}_{\delta} e_t. \quad \dots\dots\dots (4 \text{ points})$$

The objective function can be rewritten as

$$\min_{\gamma} \frac{1}{2} \frac{\sigma_e^2}{1 - \beta} (\gamma^2 + \lambda\delta^2). \quad \dots\dots\dots (6 \text{ points})$$

The FOC gives

$$\gamma^* = \frac{\lambda(1 - \beta\rho)}{\lambda(1 - \beta\rho)^2 + \kappa^2}. \quad \dots\dots\dots (9 \text{ points})$$

2. (3 points) Substituting γ^* into $x_t = \delta e_t$, we have

$$x_t = -\frac{\kappa e_t}{\lambda(1 - \beta\rho)^2 + \kappa^2}.$$